ChatGPT said:

Perfect — now that you've built the **property-focused industry risk layer**, the next step is to **turn your whole industry risk repo into a clean, reusable, portfolio-grade project** that aligns with your PD, LGD, and EL engine.

Below is a **clear, structured instruction** covering:

1. **How to restructure the repo (target architecture)**
2. **How to clean and standardise it (GitHub best practices)**
3. **How to align it with your other projects (PD / LGD / EL)**

🎯 Final Target Positioning

Rename and position your repo as:

👉 **Australian Property & Industry Risk Reference Layer**

This repo should:

- use public data (ABS, APRA, RBA)
- produce **reference tables**
- be used by:
  - PD models
  - LGD models
  - EL engine

👉 It is NOT a model — it is a **data layer**

🧱 PART 1 — Restructure the repo

Step 1 — Rename the repo (recommended)

Use one of:

> industry-risk-reference-layer

or

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> australian-property-industry-risk

Step 2 — Rebuild folder structure

Restructure your repo to:

```
industry-risk-reference-layer/
│
├─ README.md
├─ requirements.txt
├─ .gitignore
│
├─ data/
│  ├─ raw/
│  │  ├─ abs/
│  │  ├─ apra/
│  │  ├─ rba/
│  │  └─ manual/
│  │
│  ├─ processed/
│  │  ├─ property/
│  │  └─ industry/
│  │
│  └─ output/
│  ├─ region_risk/
│  │  └─ region_risk_table.csv
│  ├─ property_cycle/
│  │  └─ property_cycle_table.csv
│  ├─ arrears_environment/
│  │  └─ base_arrears_environment.csv
│  └─ downturn_overlays/
│  └─ property_downturn_overlays.csv
│
├─ notebooks/
│  ├─ 01_project_overview.ipynb
```

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```
│ ├─ 02_abs_property_data_processing.ipynb
│ ├─ 03_arrears_environment_analysis.ipynb
│ ├─ 04_region_risk_banding.ipynb
│ ├─ 05_property_cycle_banding.ipynb
│ ├─ 06_downturn_overlay_design.ipynb
│ └─ 07_final_outputs.ipynb
│
├─ src/
│ ├─ config.py
│ ├─ data_loader_abs.py
│ ├─ data_loader_apra.py
│ ├─ data_loader_rba.py
│ ├─ region_risk.py
│ ├─ property_cycle.py
│ ├─ arrears_environment.py
│ ├─ downturn_overlay.py
│ └─ utils.py
│
└─ docs/
├─ project_overview.md
├─ methodology_region_risk.md
├─ methodology_property_cycle.md
├─ methodology_arrears.md
├─ methodology_downturn.md
└─ limitations_and_assumptions.md
```

Step 3 — Separate logic cleanly

Make sure:

- **raw/** = original files (ABS, APRA, RBA)
- **processed/** = cleaned datasets
- **output/** = final reference tables

👉 Only `output/` should be used by other projects

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🧠 PART 2 — Structure the logic (very important)

Your repo should clearly produce **4 outputs**:

---

1. Region risk table

File:

> region_risk_table.csv

Purpose:

- capture regional property risk differences

Used by:

- property PD
- LGD

---

2. Property cycle / market softness table

File:

> property_cycle_table.csv

Purpose:

- identify construction cycle stage

Used by:

- development PD
- LGD
- stress testing

---

3. Base arrears environment

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File:

> base_arrears_environment.csv

Purpose:

- macro housing risk level

Used by:

- PD calibration
- EL narrative

---

4. Downturn overlays

File:

> property_downturn_overlays.csv

Purpose:

- stress scenarios

Used by:

- EL engine

🧱 PART 3 — Notebook structure (clean workflow)

Notebook 1 — Project overview

Explain:

- purpose of repo
- data sources
- outputs
- how PD/LGD/EL will use it

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Notebook 2 — ABS property data

- building approvals
- building activity
- lending indicators

Output:

- processed property dataset

Notebook 3 — Arrears environment

- summarise RBA / APRA context

Output:

- base arrears table

Notebook 4 — Region risk banding

- convert ABS signals → risk bands

Output:

- region_risk_table.csv

Notebook 5 — Property cycle banding

- define cycle stage
- define softness score

Output:

- property_cycle_table.csv

Notebook 6 — Downturn overlays

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- define stress scenarios

Output:

- property_downturn_overlays.csv

Notebook 7 — Final outputs

- export all tables cleanly

🧹 PART 4 — Clean the repo (GitHub best practices)

Step 1 — Clean notebooks

- clear outputs before commit
- remove unnecessary prints
- keep only key charts

Step 2 — Move logic to `src/`

Example:

- banding logic → `region_risk.py`
- cycle logic → `property_cycle.py`

👉 notebooks = workflow
👉 src = reusable logic

Step 3 — Remove clutter

Delete:

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- temporary files
- duplicate CSVs
- unused notebooks

Step 4 — Standardise output naming

Use consistent naming:

> region_risk_table.csv
>
> property_cycle_table.csv
>
> base_arrears_environment.csv
>
> property_downturn_overlays.csv

This aligns with:

- `pd_final.csv`
- `lgd_final.csv`
- `expected_loss.csv`

Step 5 — Improve commit history

Use logical commits:

> Restructure repo into reference-layer format
>
> Add ABS property data processing
>
> Build region risk bands
>
> Build property cycle table
>
> Add arrears environment table
>
> Add downturn overlay scenarios
>
> Rewrite README and docs

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📃 PART 5 — Rewrite README (VERY IMPORTANT)

Title

Australian Property & Industry Risk Reference Layer

Purpose

This project builds reusable risk-reference tables for property-backed lending using Australian public data.

Data sources

- ABS Building Approvals
- ABS Building Activity
- ABS Lending Indicators
- RBA housing arrears / Financial Stability Review
- APRA banking/property statistics

Outputs

- region risk table
- property cycle table
- arrears environment table
- downturn overlay table

How other projects use this

Explain:

- PD uses region + cycle

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- LGD uses region + downturn
- EL uses scenarios

Limitations

- public aggregate data only
- not borrower-level
- illustrative banding

📘 PART 6 — Add documentation

Create:

> docs/project_overview.md
> docs/methodology_region_risk.md
> docs/methodology_property_cycle.md
> docs/methodology_arrears.md
> docs/methodology_downturn.md
> docs/limitations_and_assumptions.md

🔗 PART 7 — Align with your full portfolio

After restructuring:

> Industry Risk Repo → reference tables
>
> PD Repo → uses region + cycle
> LGD Repo → uses region + downturn
> EL Engine → uses scenarios

🚀 PART 8 — What makes this strong

Most candidates:

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- build models only

You:

- build **data layer + models + portfolio engine**

👉 This is **very strong for private credit roles**

🧠 How to explain in interviews

Say:

> *I built a standalone industry and property risk reference layer using Australian public data, which feeds into my PD and LGD models as macro and regional risk adjustments, and into my Expected Loss engine for stress testing.*

✅ Final checklist

Before finishing:

- [ ] Repo renamed
- [ ] Folder structure cleaned
- [ ] 4 output tables created
- [ ] Notebooks structured clearly
- [ ] README rewritten
- [ ] Docs added
- [ ] Naming aligned with PD/LGD/EL
- [ ] Commit history improved

🔥 Final one-line summary

👉 Turn your industry risk repo into a **clean, reusable macro + property risk data layer** that feeds your PD, LGD, and EL projects.

If you want next, I can:

- write a **ready-to-paste README.md**

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- or generate **Python code for region/cycle banding logic**

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